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Review

Artificial intelligence in brain MRI analysis of Alzheimer's disease over the past 12 years: A systematic review

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ABSTRACT

Introduction: Multiple structural brain changes in Alzheimer's disease (AD) and mild cognitive impairment (MCI) have been revealed on magnetic resonance imaging (MRI). There is a fast-growing effort in applying artificial intelligence (AI) to analyze these data. Here, we review and evaluate the AI studies in brain MRI analysis with synthesis.

Methods: A systematic review of the literature, spanning the years from 2009 to 2020, was completed using the PubMed database. AI studies using MRI imaging to investigate normal aging, mild cognitive impairment, and AD-dementia were retrieved for review. Bias assessment was completed using the PROBAST criteria.

Results: 97 relevant studies were included in the review. The studies were typically focused on the classification of AD, MCI, and normal aging (71% of the reported studies) and the prediction of MCI conversion to AD (25%). The best performance was achieved by using the deep learning-based convolution neural network algorithms (weighted average accuracy 89%), in contrast to 76-86% using Logistic Regression, Support Vector Machines, and other AI methods.

Discussion: The synthesized evidence is paramount to developing sophisticated AI approaches to reliably capture and quantify multiple subtle MRI changes in the whole brain that exemplify the complexity and heterogeneity of AD and brain aging.

1. Introduction

Alzheimer's disease (AD), the most common type of late-life dementia, is multifaceted with numerous risk factors and complex expressions, comprising characteristic pathophysiological processes in the brain (Jack et al., 2018; Livingston et al., 2020). AD starts with a gradual loss of memory and cognitive function, progressing through a transient clinical stage of mild cognitive impairment (MCI) (Jack et al., 2018). Advanced age is the biggest risk factor for sporadic AD, whilst individuals age heterogeneously and have an unequal vulnerability (Au et al., 2015). Definite diagnosis of AD relies on post-mortem neuropathological gold standard of characteristic accumulation of β-amyloid plaques and neurofibrillary tangles (Alzheimer et al., 1995). Detecting AD-related brain changes at early stages is key for more effective clinical intervention and prevention of disease progression and onset (Frisoni et al., 2010; Au et al., 2015; Livingston et al., 2020). Neuroimaging modalities, in particular, magnetic resonance imaging (MRI), are promising to reveal in vivo structural brain changes. MRI-based measures of medial temporal lobe atrophy are regarded as valid markers in clinical AD diagnosis. MRI has also been uniquely used to noninvasively monitor longitudinal trajectories of cortical gray- and white- matter and small vessel abnormalities in individual patients with AD (Fox and Schott, 2004; Brickman et al., 2008; Banerjee et al., 2018; Grajauskas et al., 2019; Kaskikallio et al., 2020).

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MRI evaluation of AD is traditionally nontrivial and highly dependent on clinical experts' domain knowledge, and several rating scales have been established to aid decision-making (Pasquier et al., 1996; Wahlund et al., 2001; Chen et al., 2010; Grajauskas et al., 2018). Computer software are being developed for automated MRI assessment in support of fast segmentation and information retrieval (Bryan, 2016; Pini et al., 2016; Jiang et al., 2018; Mateos-Pérez et al., 2018). Meanwhile, barriers have stalled the translation from research to clinical implementation, including proof of generalizability of the automation to diverse disease expression and the reliability of the automated diagnosis results (Bron et al., 2015a, 2015b; Despotović et al., 2015; Pini et al., 2016; Zhavoronkov et al., 2019). The accumulated large amount of information-rich MRI data from recent multi-year and multi-center efforts such as the Alzheimer Disease Neuroimaging Initiative (ADNI; Jack et al., 2008 and the OASIS data; Marcus et al., 2007) have benefitted the development and application of more sophisticated artificial intelligence (AI) techniques, especially machine learning and deep learning on AD (see Appendix A: Technical Remarks for more details).

Despite increasing interest and accumulating research studies employing various AI approaches, a comprehensive evaluation of the characteristics, performances, and applicability of AI techniques on MRI in AD has been scarce until now (Jo et al., 2019; Elliott, 2020). In recent years, there has been a rapidly accelerating rate of new publications in this field. Synthesizing the updated literature to provide insights on the current status, challenges, and future directions has revealed a pressing need in advancing this important research field. The objectives of this paper are to evaluate the updated literature of AI studies on MRI for AD to date, assess the merits and limitations, and identify the future direction of the field in advancing AD and brain aging research. Specifically, this systematic review assessed studies whose primary purpose was to employ AI statistical models, including machine learning and deep learning, to evaluate, predict, classify, or otherwise improve the diagnosis of aging and dementia. This review focused only on studies that included brain MRI, either alone or in combination with other data, such as genetic information, other medical imaging modalities, demographic information, or cognitive test scores.

To help the readers better keep up with the AI methods referred to in this review, we included an Appendix of Technical Remarks to summarize the involved predominant algorithms regarding key concepts, terminologies, and application potentials (see Appendix A. Technical Remarks: Sections A.4 to A.7).

2. Methods

This paper has been prepared according to the standard PRISMA (preferred reporting items for systematic reviews and meta-analyses) guidelines (Page et al., 2021; the PRISMA 2020 Checklist is presented in the Appendix file).

2.1. Databases and coverage of search

We searched PubMed, a comprehensive biomedical and life sciences literature database maintained by the US National Library of Medicine that contains more than 30 million published works. We targeted all the original research publications reporting human brain MRI research on aging and AD with AI applications, from January 1st, 2009, until December 31st, 2020 (i.e., spanning 12 years). The search was completed on January 13th, 2021. PubMed searches the title, authors, publication date, keywords, and abstract fields. If the search terms only appear in the body of the text or figure captions of certain articles, they may not show up in the search results. PubMed search results were supplemented with some research articles from backward reference searching that met the appropriate criteria outlined in 2.3.

2.2. Search terms

The search used three sets of inclusion terms and a few exclusion terms; case insensitive. The wildcard symbol "* " was used to signify the inclusion of multiple possible suffixes whenever appropriate. Terms within a set were combined using OR; different inclusion term sets were combined using AND; NOT were used to indicate the exclusion terms.

Inclusion Term Set 1. *:* "Magnetic Resonance Imaging" OR "MRI" OR "T1" OR "T1WI" OR "T2" OR "T2WI" OR "Structural MRI" OR "Anatomical MRI" OR "Diffusion Tensor Imaging" OR "DTI" OR "Diffusion weighted imaging" OR "DWI" OR "Perfusion MRI" OR "PWI" OR "fMRI" OR "functional MRI".

Inclusion Term Set 2. *:* "Alzheimer's Disease" OR "AD" OR "Aging" OR "Dementia" OR "Mild Cognitive Impairment" OR "MCI" OR "Amnestic MCI" OR "Non*Amnestic MCI" or "Mixed AD*VAD" OR "Vascular Dementia".

Inclusion Term Set 3. *:* "Artificial Intelligence" OR "Machine Learning" OR "Deep Learning" OR "Neural Network" OR "Computer Assisted Diagnosis" OR "Medical Informatics" OR "RNN" OR "CNN" OR "Random Forest" OR "SVM" OR "Support vector machine".

Exclusion Terms. *:* "schizophrenia" OR "depression" OR "major depressive disorder".

The search yielded a total of 1463 article entries initially (Fig. 1).

2.3. Review criteria

The initial entries were filtered to include only those that investigated adults aged 19^{+} years. Additional restrictions granted full-text, peer-reviewed original research reported in English (Fig. 1).

Following the initial title and abstract reading, publications that met the following exclusion criteria were removed from the review process (Fig. 1).

1. Analysis on developing brains (dataset included subjects <19 years of age)
2. Studies of animals or ex-vivo tissue.

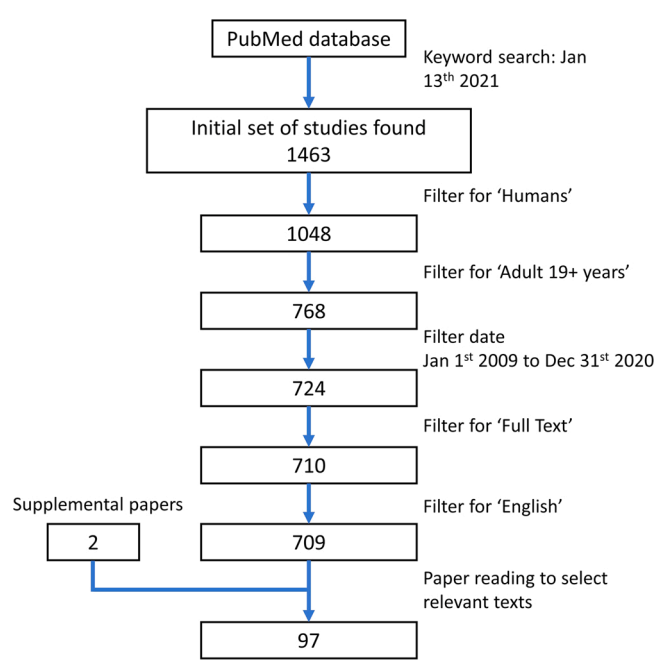

Fig. 1. PubMed database paper search schematic.

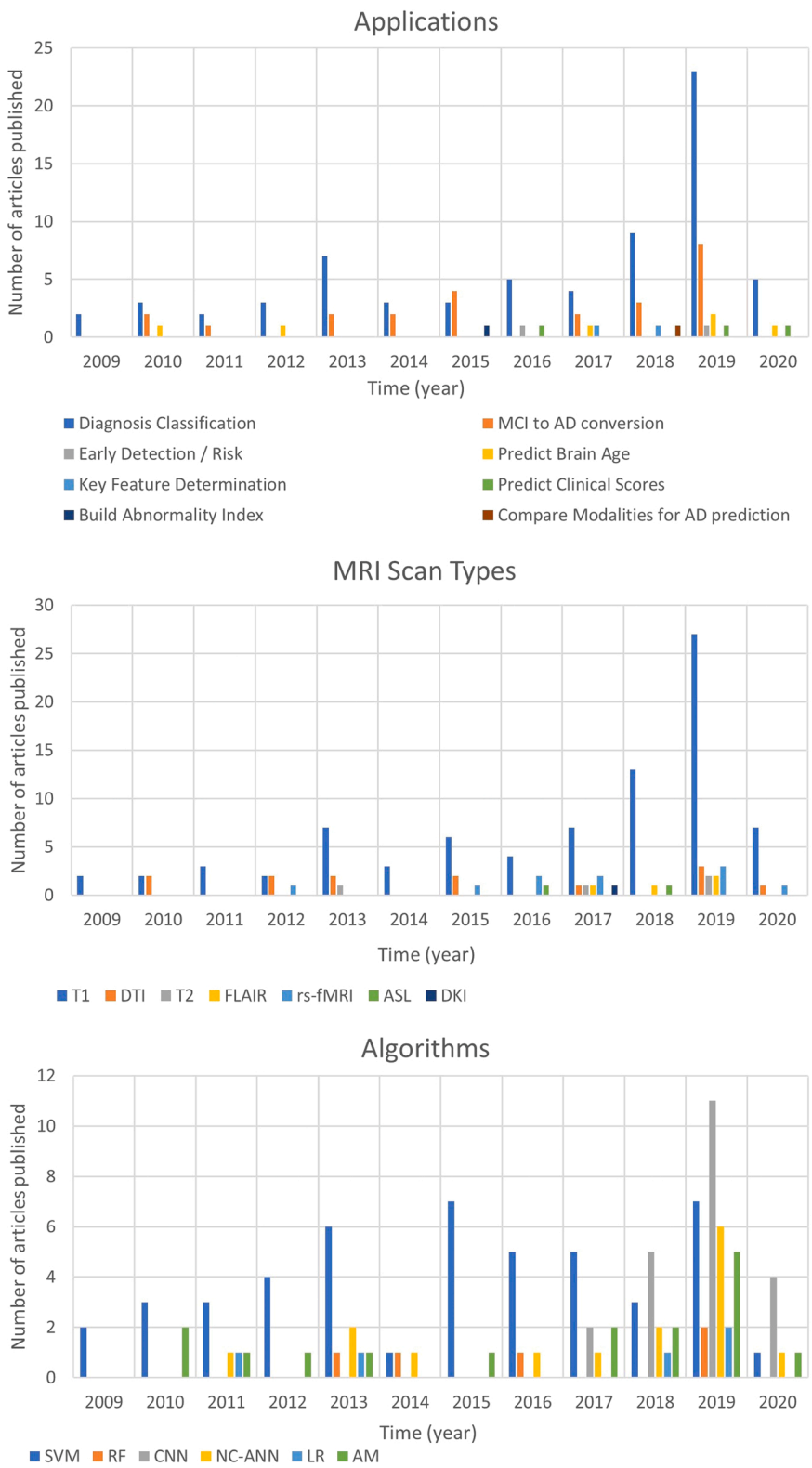

Fig. 2. Annual trends in reviewed publications from 2009 to 2020. Panel A: Yearly totals for AI method applications. Panel B: Yearly totals for MRI Scan types used for AI algorithm data. Panel C: Yearly totals for major AI algorithm types employed.

3. Studies including only medical imaging methods other than MRI (e.g., Computed tomography, PET)
4. MRI images not of the brain (e.g., spinal cord imaging, cardiac imaging)
5. Studies that did not focus on brain aging or dementia (e.g., Schizophrenia, depression, traumatic brain injury, etc.)
6. Reviews, commentaries, surveys, conference abstracts, or anything where a peer-reviewed, full-length original research article could not be retrieved.

7. Studies that did not include a fully automated AI or machine learning technique. Data preprocessing or brain parcellation, or other similar feature selection using validated software or domain experts were included.
8. Studies reported in a language other than English
9. Studies that did not present their methods in sufficient detail to determine if they met the inclusion/exclusion criteria.

The above operations yielded a final set of 97 articles for in-depth full-text study, each reported original research using any AI methods analyzing any type of brain MRI in studying AD and MCI and related brain aging (Fig. 1).

2.4. Data analysis

Each of the eligible articles was evaluated in detail, and data were extracted and assessed independently by two separate authors (Fig. 1; Tables 1–3).

Dataset information including data origin (e.g., database) and participant information, scan information including MRI scan types and field strength, study methods including algorithm model and performance metrics were collected.

The performance measures collected were accuracy (which was reported by the majority of the studies), AUC, sensitivity, specificity for classification models. If these were not present in a study, other report metrics were recorded such as the adjusted or weighted R^2 for regression models. When multiple results were reported for the same algorithm and application, the best metrics of the testing/validation data were recorded.

Tables 2A to 2F present this data organized primarily by AI algorithm type. The forest plots in Appendix B; Fig. B1 – B6, present studies reporting the AUC and accuracy and are separated based on algorithm application and algorithm type. The forest plots were generated in R and the 95% confidence intervals were calculated using Wilson's interval (Wilson, 1927).

Table 1
Total number of research articles under review, by the type of MRI scan, AD application, and AI algorithm.

	CATEGORY	Total number of studies[a]
MRI Scan Type	T1WI	83
	DTI/DWI	13
	T2WI	4
	FLAIR	4
	fMRI (resting-state BOLD)	10
	PWI (ASL)	1
	DKI	1
AD Application	Diagnosis Classification	69
	MCI to AD conversion	24
	Predict Brain Age	6
	Early Detection	2
	Key Feature Determination	2
	Predict Clinical Scores	3
	Build Abnormality Index	1
	Predict Cognitive Decline	1
AI Algorithm	Support Vector Machine	47
	Non-Convolutional Artificial Neural Network	15
	Convolutional Neural Network	22
	Random Forest	5
	Logistic Regression	5
	Additional Methods	16

The 'Additional Methods' category encompassed several other algorithms from a smaller number of applications.
AD – Alzheimer's Disease. NC-ANN – Non-Convolutional Artificial Neural Network. ASL – Arterial Spin Labelling. CNN – Convolution Neural Network. DKI – Diffusion Kurtosis Imaging. DTI – Diffusion Tensor Imaging. DWI – Diffusion Weighted Imaging. FLAIR – Fluid-Attenuated Inversion Recovery. fMRI – functional Magnetic Resonance Imaging. MCI – Mild Cognitive Impairment. PWI – Perfusion Weighted Imaging. SVM – Support Vector Machine. T1WI – T1 weighted imaging. T2WI – T2 weighted imaging.
[a] Papers that satisfied multiple classification characteristics are shown with multiple entries.

2.5. Risk of bias assessment

The risk of bias was assessed using the PROBAST (prediction model risk of bias assessment tool) criteria (Wolff et al., 2019). PROBAST employs 20 signaling questions across four domains: Participants, Predictors, Outcome, Analysis. Questions pertaining to the domains relevant to the studies were included in the analysis, except for predictors that are less relevant for AI on MRI applications. Exact signaling question inclusion and exclusion can be found in Appendix C. Additionally, an applicability analysis was not assessed as a specific therapeutic question is not posed for the reviewed studies.

Each study was reviewed and assessed independently by at least two authors. Any discrepancies (occurred in ~5% of all the incidences) were resolved through consensus or by a third reviewer.

3. Results

Table 1 summarizes the number of publications, respectively organized by the type of MRI data, AD-application, and AI algorithm. Table 2 (A through F) presents a detailed elaboration of each article associated with the common algorithms; support vector machines (SVM), non-convolutional artificial neural network (NC-ANN), convolutional neural networks (CNN), logistic regression (LR), and random forest (RF) (see Appendix A: Technical Remarks for the description of the algorithms). Studies comparing modified applications (e.g., input features, cross-validation techniques, or class delineation) of the algorithms were also included, listing the best-reported results in the original article. All other algorithms, each of which involved just one or two studies, were included under "Additional Methods" (Tables 1, 2). Performances comparing different algorithms are listed and summarized in Appendix B.

3.1. Support vector machines

SVMs are machine-learning models that determine optimal decision boundaries for classifying data (Appendix A.4). Table 2A lists all the studies using an SVM. Highlights from reviewed articles employing an SVM are reported below.

Magnin et al. (2009) automated the classification of AD vs HC based on gray matter characteristics and achieved high accuracy (94%), sensitivity (96%), and specificity (91%). Gerardin et al. (2009) modeled the hippocampal shape features for AD vs. HC, and 83% for MCI vs. HC. Plant et al. (2010) derived a pattern-matching index by combining an SVM with statistical methods for discriminating AD and HC, and in predicting 2.5 year MCI to AD conversion. Costafreda et al. (2011) reported 80% accuracy for predicting 1-year AD conversion based on automated hippocampal morphology using over a hundred MCI cases. Yang et al. (2011) reported an ICA based method to optimize the classification of AD, MCI, and HC using an SVM and achieved 86% accuracy on a secondary dataset. Additional studies examined the accuracy for AD, MCI, and HC classifications using different subcortical and cortical features (Escudero et al., 2011; Diciotti et al., 2012; Yang et al., 2013).

Cuingnet et al. (2013) proposed spatially regularizing the SVM parameters in handling 3D anatomical and 2D surface information and reported a classifier function of 91% accuracy with a framework mapping AD associated regional atrophy. Bron et al. (2015) tested gray matter feature selection and showed slightly improved classification performance for stable MCI. Klöppel et al., (2015) differentiated AD, FTD, LBD, and HC and tested the model with <80% AUC for multi-class separation. Retico et al. (2014) showed a higher AUC for AD and HC classification when whole-brain gray matter was applied. Proceeding

Table 2
List of the research articles under review, presented by AI techniques (Panels A, B, C, D, E, F).

Table 2A Studies Using Support Vector Machines

Year	First Author	Scan Type	Application	Field Strength	Dataset	Participants	Result			
							Accuracy	AUC	Specificity	Sensitivity
2009	*Magnin, B.*	T1	Diagnosis classification[‡]	1.5	Local	HC: 22 AD: 16	95		97	92
2009	*Gerardin, E.*	T1	Diagnosis classification[‡]	1.5	Local	HC: 25 MCI: 23 AD: 23	94		92	96
2010	*Haller, S.*	DTI	Diagnosis classification[^]	3 T	Local	HC: 35 MCI: 67	91		90	90
			MCI to AD conversion				98		100	98
2010	*Oliveira, P.P. Jr.*	T1	Diagnosis classification[‡]	1.5 T	Local	HC: 20 AD: 14	88		85	93
2010	*Plant, C.*	T1	Diagnosis classification[^]	1.5 T	Local	HC: 18 MCI: 24 AD: 32	98		100	96
			MCI to AD conversion				96		100	89
2011	*Costafreda, SG.*	T1	MCI to AD conversion	1.5 T	ADNI	HC: 88 MCI: 103 AD: 71	80		80	77
2011	*Escudero, J.*	T1	Diagnosis classification[‡]	1.5 T	ADNI	HC: 180 MCI: 222 AD: 122	89	88		
2011	*Yang, W.*	T1	Diagnosis classification[‡]	1.5 T	ADNI, OASIS	HC: 236 MCI: 410 AD: 202	86		81	91
2012	*Cui, Y.*	T1, DTI	Diagnosis classification[^]	3 T	Sydney MAS	HC: 204 MCI: 79	71	70	78	52
2012	*Diciotti, S.*	T1	Diagnosis classification[‡]	1.5 T	Local	HC: 29 MCI: 30 AD: 21	86		90	82
2012	*Meier, T.B.*	rs-fMRI	Predict brain age	3 T	ICBM	18–35 yrs: 26 55–85 yrs: 26	84			
2012	*O'Dwyer, L.*	DTI	Diagnosis classification[^]	3 T	Local	HC: 40 MCI: 33			93	93
2013	*Aguilar, C.*	T1, T2	MCI to AD conversion	1.5 T	AddNeuroMed	HC: 110 MCI: 119 AD: 116	88	92	90	86
2013	*Cuingnet, R.*	T1	Diagnosis classification[‡]	1.5 T	ADNI	HC: 162 AD: 137	91		93	88
2013	*Dukart, J.*	T1, FDG-PET	Diagnosis classification[‡]	1.5 T	ADNI, Leipzig Cohort	HC: 41 AD: 49	90		88	92
2013	*Dyrba, M.*	DTI, T1	Diagnosis classification[‡]	1.5/3 T	ESDS	HC: 143 AD: 137	89		91	87
2013	*Haller, S.*	DTI	Diagnosis classification[ε]	3 T	Local	MCI: 66	100			
2013	*Yang, S.T.*	T1	Diagnosis classification[‡]	3 T	Local	HC: 17 MCI: 18 AD: 17	83		83	88
2014	*Apostolova, LG.*	T1	Diagnosis classification[‡]	1.5 T	ADNI	HC: 111 MCI: 182 AD: 95	85	88		
2015	*Andrade de Oliveira, A.*	T1	Build abnormality index	1.5 T	ADNI	HC: 228 MCI: 395 AD: 191	84			
2015	*Bron, E.E.*	T1	Diagnosis classification[‡]	1.5 T	ADNI	HC: 162 MCI: 210 AD: 137		92		
			MCI to AD conversion					61		
2015a	*Dyrba, M.*	DTI	Diagnosis classification[^]	1.5/3 T	EDSD	HC: 25 MCI: 95	77		80	73
2015b	*Dyrba, M.*	rs-fMRI, DTI, T1	Diagnosis classification[‡]	3 T	DZNE	HC: 25 AD: 28	85	89	92	79
2015	*Kloppel, S.*	T1	MCI to AD conversion	1.5/3 T	Local	MCI: 28 AD: 122 FTD: 12		97		
2015	*Moradi, E.*	T1	MCI to AD conversion	1.5 T	ADNI	HC: 231 MCI: 394 AD: 200	82	90	74	87
2015	*Retico, A.*	T1	MCI to AD conversion	1.5 T	ADNI	HC: 189 MCI: 302 AD: 144	80	87	80	80
2016	*Collij, L.E.*	ASL	Diagnosis classification[λ]	3 T	Local	HC: 26 MCI:60 AD: 100 SCD:100	89	93		
2016	*Khazaee, A.*	rs-fMRI	Diagnosis classification[σ]	3 T	ADNI	HC: 45 AD: 34 MCI: 89	98			
2016	*Linn, K.A.*	T1	Diagnosis classification[‡]	1.5 T	ADNI	HC: 230 AD: 200	87	89	88	85
2016	*Long, Z.*	rs-fMRI	Early detection	3 T	Local	HC: 33 MCI: 29	97	97	100	93
2016	*Moller, C.*	T1	Diagnosis classification[‡]	3 T	Local	HC: 94 AD: 84 bvFTD: 51	85	95	87	83
2017	*Chen, Y.*	T1, DTI, DKI	Diagnosis classification[‡]	3 T	Local	HC: 26 AD: 27	96	99	93	100
2017	*Hojjati, S.H.*	rs-fMRI	MCI to AD conversion	1.5 T	ADNI	MCI: 80	91	95	90	83
2017	*Lama, R.K.*	T1	Diagnosis classification[‡]	3 T	ADNI	HC: 70 MCI: 74 AD: 70	80	78	79	83
2017	*Long, X.*	T1	MCI to AD conversion	1.5/3 T	ADNI	HC: 135 MCI: 227 AD: 65	89	93	91	86

(*continued on next page*)

Table 2 (*continued*)

Table 2A Studies Using Support Vector Machines

Year	First Author	Scan Type	Application	Field Strength	Dataset	Participants	Result			
							Accuracy	AUC	Specificity	Sensitivity
2017	*Yang, W*	T1	Diagnosis classification‡	1.5 T	ADNI	HC: 150 MCI: 98 AD: 70	98		97	99
2018	*Ferreira, L.K.*	T1, rCBF-SPECT, FDG-PET	Compare Modalities for AD prediction	1.5 T	Local	HC: 18 AD: 20	74	82		
2018	*Grassi, M.*	T1	MCI to AD conversion	Un-reported	Local	HC: 136 preMCI: 94 MCI: 90	87	91	88	87
2018	*Sun, Z.*	T1	Diagnosis classification‡	1.5 T	ADNI	HC: 162 MCI: 210 AD: 137	93	96	93	91
			MCI to AD conversion				65	68	64	68
2019	*Peraza, L.*	DTI	Diagnosis classification‡	1.5 T	ADNI, NKI	HC: 91 AD: 39	89		99	79
2019	*Seo, K.*	T1	Diagnosis classification‡	1.5 T	ADNI	HC: 177 MCI: 277 AD: 108	92			
2019	*Sheng, J.*	rs-fMRI, T1, T2	Diagnosis classification‡	1.5 T	ADNI	HC: 24 MCI: 48 AD: 24	96			
2019	*Tam, A.*	T1	MCI to AD conversion	1.5/3 T	ADNI	HC: 446 MCI: 751 AD: 295	85	87	97	47
2019	*Yamashita, A.*	T1, FDG-PET	Diagnosis classification‡	1.5/3 T	ADNI	HC: 222AD: 181	90	94	93	86
2019	*Zhao, W.*	T1	Diagnosis classificationˆ	3 T	Local	HC: 48 MCI: 44 SCD: 40	96	99		
2019	*Zheng, Y.*	T1, T2 FLAIR	Diagnosis classificationΨ	1.5 T	Local	AD: 58 VaD: 35	84	86		
2020	*Gao, Y.*	T1, rs-fMRI	Diagnosis classification‡	1.5/3 T	ADNI	HC: 136 MCI: 166 SCD: 46 AD: 35		87	81	83

Abbreviations: AD – Alzheimer's Disease. ADNI – Alzheimer's Disease Neuroimaging Initiative. ASL – Arterial Spin Labelling. bvFTD – behavioral variant Frontotemporal Dementia. DKI – Diffusion Kurtosis Imaging. DTI – Diffusion Tensor Imaging. DZNE – German Center for Neurodegenerative Diseases. EDSD – European DTI Study on Dementia. FDG-PET – Fluorodeoxyglucose - Positron Emission Tomography. FLAIR – Fluid-Attenuated Inversion Recovery. FTD – Frontotemporal Dementia. HC – Healthy Control. MCI – Mild Cognitive Impairment. NHIS-IH – National Health Insurance Service-Ilsan Hospital. NKI – Nathan Kline Institute-Rockland Sample. rCBF-SPECT – regional Cerebral Blood Flow Single Photon Emission Computed Tomography. rs-fMRI – resting state fMRI. SCD – Subjective Cognitive Decline. Sydney MAS – Sydney Memory and Aging Study. VaD – Vascular Dementia.

Ψ Diagnosis classification representing AD vs. VaD

ε Diagnosis classification representing single domain amnestic MCI vs. single domain frontal MCI

λ Diagnosis classification representing AD vs SCD

σ Diagnosis classification representing AD vs. HC & MCI

ˆ Diagnosis classification representing MCI vs. HC classification.

‡ Diagnosis classification representing AD vs. HC classification.

Table 2B. Studies Using Non-Convolutional Artificial Neural Networks

Year	**First Author**	**Scan Type**	**Application**	**Field Strength**	**Dataset**	**Participants**	**Result**				**Technical Details**
							Accuracy	**AUC**	**Specificity**	**Sensitivity**	
2011	*Escudero, J.*	T1	Diagnosis classification‡	1.5 T	ADNI	HC: 180 MCI: 222 AD: 122	88	87			RBF
2013	*Aguilar, C.*	T1, T2	MCI to AD conversion	1.5 T	AddNeuroMed	HC: 110 MCI: 119 AD: 116	87	91	90	85	Multilayer Perceptron
2013	*Yang, S.T.*	T1	Diagnosis classification‡	3 T	Local	HC: 17 MCI: 18 AD: 17	88		88	88	SVM + PSO
2014	*Suk, H.-I.*	T1, FDG-PET	Diagnosis classification‡	1.5 T	ADNI	HC: 101 MCI: 204 AD: 93	95	99	95	95	DBM
			MCI to AD conversion				76	75	95	48	
2016	*Cárdenas-Peña, D.*	T1	Diagnosis classificationα	1.5 T	ADNI	HC: 655 MCI: 825 AD: 513	71	85			Feed Forward Network
2017	*Lama, R.K.*	T1	Diagnosis classification‡	3 T	ADNI	HC: 70 MCI: 74 AD: 70	78		84	69	ELM
2018	*Amoroso, N.*	T1	Diagnosis classification‡	1.5 T	ADNI	HC: 81 MCI: 48 AD: 86	91	97	88	98	Salient Skeleton Network
2018	*Ning, K.*	T1	Diagnosis classification‡	1.5/3 T	ADNI	HC: 225 MCI: 358 AD: 138		95			Feed Forward Network
2019	*Cohen, D.S.*	T1	Diagnosis classificationα	1.5 T	ADNI	HC: 352 MCI: 531 AD: 334	87				Feed Forward Network
2019	*Duraisamy, B.*	T1	Diagnosis classification‡	3 T	ADNI, Bordex-3 city	HC: 183 MCI: 210 AD: 153	98		86	80	WPNN
2019	*Ju, R.*	rs-fMRI	Diagnosis classificationˆ	3 T	ADNI	HC: 79 MCI: 91	86	92	81	92	Auto-encoder
2019	*Mehdipour Ghazi, M.*	T1	Diagnosis classification‡	1.5 T	ADNI	HC: 2745 MCI: 4058 AD: 2108		90			LSTM (RNN based)
2019	*Nguyen, D.T.*	rs-fMRI	Diagnosis classification‡	3 T	ADNI, Local	HC: 183 MCI: 163 AD: 114	99		98	100	ELM

(*continued on next page*)

Table 2 (*continued*)

Table 2B. Studies Using Non-Convolutional Artificial Neural Networks

Year	First Author	Scan Type	Application	Field Strength	Dataset	Participants	Result				Technical Details
							Accuracy	AUC	Specificity	Sensitivity	
2019	*Zhang, F.*	T1	Diagnosis classification‡	1.5 T	ADNI	HC: 94 AD: 58	96	93	89	98	ELM
2020	*Mendoza-Leon, R*	T1	Diagnosis classification‡	1.5 T	OASIS	HC: 87 AD: 87	90	92	85	95	Auto-encoder

Abbreviations: AD – Alzheimer's Disease. ADNI – Alzheimer's Disease Neuroimaging Initiative. DBM – Deep Boltzmann Machine. ELM – Extreme Learning Machine. HC – Healthy Control. LSTM – Long Short Term Memory network. MCI – Mild Cognitive Impairment. OASIS – Open Access Series of Imaging Studies. PSO – Particle Swarm Optimization. RBF – Radial Basis Function. RNN – Recurrent Neural Network. rs-fMRI – resting state fMRI. SVM – Support Vector Machine. SCZ – Schizophrenia. WPNN – Weighted Probabilistic Neural Network.

α Diagnosis classification representing multi-class HC, MCI, AD

ˆ Diagnosis classification representing MCI vs. HC classification.

‡ Diagnosis classification representing AD vs. HC classification.

Table 2C. Studies Using Convolutional Neural Network

Year	First Author	Scan Type	Application	Field Strength	Dataset	Participants	Result				Technical Details
							Accuracy	AUC	Specificity	Sensitivity	
2017	*Cole, J.*	T1	Predict brain age	1.5/3 T	BAHC	HC: 2001					R2 = 0.92
2017	*Ghafoorian, M.*	T1	Identify lacunes	1.5 T	RUNDMC, FUTURE	SVD/Stroke: 1075				97	
2018	*Islam, J.*	T1	Diagnosis classificationβ	1.5 T	OASIS	AD: 416	93			93	
2019	*Cohen, D.S.*	T1	Diagnosis classificationα	1.5 T	ADNI	HC: 352 MCI: 531 AD: 334	88				
2019	*Kruthika, K. R.*	T1, FDG-PET	Diagnosis classification‡	1.5 T	ADNI	HC: 605 MCI: 991 AD 345	98				CNN + Autoencoder
2018	*Li, F.*	T1	Diagnosis classification‡	1.5/3 T	ADNI	HC: 229 MCI: 403 AD: 199	90	92			
2018	*Lin, W.*	T1	MCI to AD conversion	1.5/3 T	ADNI	HC: 229 MCI: 401 AD: 188	80	86	75	84	
2018	*Liu, M.*	T1, FDG-PET	Diagnosis classification‡	1.5 T	ADNI	HC: 100 MCI: 204 AD: 93	93	96	94	93	
2018	*Wang, S.H.*	T1	Diagnosis classification‡	1.5 T	OASIS, Local	HC: 98 AD: 98	98		97	98	
2019	*Basaia, S.*	T1	Diagnosis classification‡	3 T	ADNI	HC: 407 MCI: 813 AD: 418	99		99	99	
			MCI to AD conversion				75		75	75	
2019	*Basheera, S.*	T2	Diagnosis classification‡	1.5 T	ADNI	HC: 117 MCI: 112 AD: 120	100	100	100	100	
2019	*Cui, R.*	T1	Diagnosis classification‡	1.5/3 T	ADNI	HC: 229 MCI: 403 AD: 198	91	93	95	87	CNN + RNN
			MCI to AD conversion				72	73	76	65	
2019	*Huang, Y.*	T1, FDG-PET	Diagnosis classification‡	1.5 T	ADNI	HC: 731 MCI: 767 AD: 647	90	91	89	91	
2019	*Jain, R.*	T1	Diagnosis classification‡	1.5 T	ADNI	HC: 50 MCI: 50 AD: 50	99				
2019	*Jonsson, B.A.*	T1	Predict Brain Age	1.5 T	IXI, UK Biobank	HC: 18642					R2 = 0.83
2019	*Li, H.*	T1	MCI to AD conversion	1.5/3 T	ADNI, AIBL	MCI: 2146		79			
2019	*Oh, K.*	T1	Diagnosis classification‡	1.5 T	ADNI	HC: 230 MCI: 267 AD: 198	87	91	85	89	
			MCI to AD conversion				74	79	71	77	
2019	*Spasov, S.*	T1	MCI to AD conversion	1.5 T	ADNI	HC: 184 MCI: 409 AD: 192	86	93	84	88	
2020	*Bashyam, V.*	T1	Predict Brain Age	Un-reported	LifespanCN	HC: 11729					R = 0.98
2020	*Pan, D*	T1	Diagnosis classification‡	1.5 T	ADNI	HC: 262 MCI: 288 AD: 237	84	92			
2020	*Marzban, E*	T1, DTI	Diagnosis classification‡	1.5 T	ADNI	HC: 185 MCI: 106 AD: 115	94	94	94	93	
2020	*Mehmood, A.*	T1	Diagnosis classificationβ	1.5 T	OASIS	HC: 334 MCI: 384 AD: 46	98				

Abbreviations: AD – Alzheimer's Disease. AIBLE – Australian Imaging Biomarkers and Lifestyle Study of Aging. BAHC – Brain-Age Healthy Control dataset. CNN – Convolution Neural Network. FDG-PET – Fluorodeoxyglucose - Positron Emission Tomography. FUTURE – Follow-Up of transient ischemic attack and stroke patients and Unelucidated Risk factor

(*continued on next page*)

Table 2 (*continued*)

Table 2C. Studies Using Convolutional Neural Network

Year	First Author	Scan Type	Application	Field Strength	Dataset	Participants	Result				Technical Details
							Accuracy	AUC	Specificity	Sensitivity	

Evaluation study. HC – Healthy Control. IXI – Information eXtraction from Images. MCI – Mild Cognitive Impairment. OASIS – Open Access Series of Imaging Studies. RNN – Recurrent Neural Network. RUNDMC – Radboud University Nijmegen Diffusion tensor and Magnetic resonance imaging Cohort. SVD – Small Vessel Disease.
[α] Diagnosis classification representing multi-class HC, MCI, AD
[β] Diagnosis classification representing multi-class HC, very mild AD, mild AD, and moderate AD
[‡] Diagnosis classification representing AD vs. HC classification.

Table 2D. Studies Using Random Forest

Year	First Author	Scan Type	Application	Field Strength	Dataset	Participants	Result				Technical Details
							Accuracy	AUC	Specificity	Sensitivity	
2013	*Gray, K.R.*	T1, FDG-PET	Diagnosis classification[‡]	1.5 T	ADNI	HC: 35 MCI: 75 AD: 37	89	89	90	88	
2014	*Lebedev, A.V.*	T1	Diagnosis classification[‡]	1.5 T	ADNI, AddNeuroMed	HC: 325 MCI: 279	90	94	92	89	
			MCI to AD conversion			AD: 292	82		81	83	
2016	*Huang, L.*	T1	Predict Clinical Scores	1.5 T	ADNI	HC: 226 MCI: 393 AD: 186					w$R = 0.90$
2019	*Moore, P. J.*	T1	Diagnosis classification[α]	1.5 T	ADNI	HC: 195 MCI: 150 AD: 72	73	82			
2019	*Wang, Y.*	T1, DTI, T2 FLAIR	Diagnosis classification[ω]	1.5/3 T	NHIS-IH, ADNI	HC: 107 MCI: 122 AD: 158		99			

Abbreviations: AD – Alzheimer's Disease. ADNI – Alzheimer's Disease Neuroimaging Initiative. DTI – Diffusion Tensor Imaging. FDG-PET – Fluorodeoxyglucose - Positron Emission Tomography. FLAIR – Fluid-Attenuated Inversion Recovery. HC – Healthy Control. MCI – Mild Cognitive Impairment. NHIS-IH – National Health Insurance Service-Ilsan Hospital.
[α] Diagnosis classification representing multi-class HC, MCI, AD
[ω] Diagnosis classification representing AD vs. MCI
[‡] Diagnosis classification representing AD vs. HC classification.

Table 2E. Studies Using Logistic Regression

Year	First Author	Scan Type	Application	Field Strength	Dataset	Participants	Result			
							Accuracy	AUC	Specificity	Sensitivity
2011	*Escudero, J.*	T1	Diagnosis classification[‡]	1.5 T	ADNI	HC: 180 MCI: 222 AD: 122	86	92		
2013	*Casanova, R.*	T1	Diagnosis classification[‡]	1.5 T	ADNI	HC: 188 MCI: 335 AD: 171	87		89	84
			MCI to AD conversion				62		76	46
2018	*Casanova, R.*	T1	Diagnosis classification[‡]	1.5 T	ADNI, WHIMS	HC: 188 MCI:1365 AD: 171		94		
2019	*Moscoso, A.*	T1	MCI to AD conversion	1.5/3 T	ADNI	HC: 230 MCI: 248 AD: 230		84	71	78
2019	*Wang, Y.*	T1, DTI, T2 FLAIR	MCI to AD conversion	1.5/3 T	NHIS-IH, ADNI	HC: 107 MCI: 122 AD: 158	69	79	69	79

Abbreviations: AD – Alzheimer's Disease. ADNI – Alzheimer's Disease Neuroimaging Initiative. DTI – Diffusion Tensor Imaging. FLAIR – Fluid-Attenuated Inversion Recovery. HC – Healthy Control. MCI – Mild Cognitive Impairment. NHIS-IH – National Health Insurance Service-Ilsan Hospital. WHIMS – Women's Health Initiative Memory Study.
[‡] Diagnosis classification representing AD vs. HC classification.

Table 2 F. Studies Using Additional Methods

Year	First Author	Scan Type	Application	Field Strength	Dataset	Participants	Result				Technical Details
							Accuracy	AUC	Specificity	Sensitivity	
2010	*Plant, C.*	T1	Diagnosis classification[‡]	1.5 T	Local	HC: 18 MCI: 24 AD: 32	92		89	94	Bayes, VFI*
2010	*Robinson, E. C.*	DTI	Predict Brain Age	3 T	Local	20–30 yrs: 40 60–90 yrs: 54	87				LDA
2011	*Escudero, J.*	T1	Diagnosis classification[‡]	1.5 T	ADNI	HC: 180 MCI: 222 AD: 122	84	83			C4.5
2012	*Diciotti, S.*	T1	Diagnosis classification[‡]	1.5 T	Local	HC: 29 MCI: 30 AD: 21	78		83	72	Naïve Bayes
2013	*Aguilar, C.*	T1, T2	Diagnosis classification[‡]	1.5 T	AddNeuroMed	HC: 110 MCI: 119 AD: 116	86	92	92	80	OPLS* , Trees;
2015	*Moradi, E.*	T1	MCI to AD conversion	1.5 T	ADNI	HC: 231 MCI: 394 AD: 200	75	77	52	89	LDS
2017	*De Marco, M.*	rs-fMRI, T1, T2, FLAIR	Diagnosis classification^	1.5 T	Local	HC: 50 MCI: 50	85				LDA, QDA*

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Table 2 (*continued*)

Table 2 F. Studies Using Additional Methods

Year	First Author	Scan Type	Application	Field Strength	Dataset	Participants	Result				Technical Details
							Accuracy	AUC	Specificity	Sensitivity	
2017	*Lama, R.K.*	T1	Diagnosis classification‡	3 T	ADNI	HC: 70 MCI: 74 AD: 70	74		65	87	IVM
2018	*Jiang, J.*	T1, T2 FLAIR	WMHI segmentation pipeline	1.5/3 T	Sydney MAS, OATS	HC: 166					kNN ICC = 99
2018	*Lee, J.S.*	T1	Diagnosis classification‡	3 T	Local	HC: 869 MCI: 79 AD: 500	91		93	87	LDA
2019	*Beheshti, I.*	T1	Predict Brain Age	1.5/3/4 T	PPMI, ICBM, MindBoggle-101, FBIRN, Kirby-21, IXI	HC: 178					P-BGDF R2 = 0.94
2019	*Bouts, M.J. R.J*	T1, DTI	Diagnosis classification^	1.5/3 T	Rotterdam study	HC: 790 MCI: 48 AD: 77	65	62	66	59	Elastic net regression
2019	*Jiang, P.*	T1	Predict Cognitive Scores	1.5 T	ADNI	HC: 197 MCI: 295 AD: 130					CSL MSE = 0.57
2019	*Kim, J.P.*	T1, FDG-PET	Diagnosis classification+	3 T	Local	HC: 146 AD: 50 FTD: 143	92	96	88	97	LDA
2019	*Lisowska, A.*	T1	Early detection	1.5 T	ADNI	HC: 42 MCI: 42	73	76	79	67	GGL-SCCA
2020	*Franzmeier, N.*	T1, FDG-PET	Predict Cognitive Decline	3 T	ADNI, DIAN	HC: 49 MCI: 391					SVR R2 = 0.53

Abbreviations: AD – Alzheimer's Disease. ADNI – Alzheimer's Disease Neuroimaging Initiative. C4.5 – Decision Tree Algorithm. CSL – Correlation-Aware Sparse and Low-Rank Constrained Multi-Task Learning. DIAN – Dominantly Inherited Alzheimer Network. DTI – Diffusion Tensor Imaging. FBIRN – Functional Biomedical Informatics Research Network. FDG-PET – Fluorodeoxyglucose - Positron Emission Tomography. FLAIR – Fluid-Attenuated Inversion Recovery. FTD – Frontotemporal Dementia. GGL-SCCA – graph-guided pairwise group lasso based sparse canonical correlation analysis model. HC – Healthy Control. ICBM – International Consortium for Brain Mapping. ICC – Intraclass Correlation Coefficient. IVM – Import vector machine. IXI – Information eXtraction from Images. kNN – k Nearest Neighbor. LDA – Linear Discriminant Analysis. LDS – Low Density Separation. MCI – Mild Cognitive Impairment. MSE – Mean Squared Error. OATS – Older Australian Twins Study. OPLS – Orthogonal Projections to Latent Structures. P-BGDF – Patch-based grading-driven framework. PPMI – Parkinson's Progression Markers Initiative. QDA – Quadratic Discriminant Analysis. RBF – Radial Basis Function. rs-fMRI – resting state fMRI. SVR – Support Vector Regression. Sydney MAS – Sydney Memory and Aging Study. VFI – Voting Feature Intervals. WMHI – White Matter Hyperintensities.

* Algorithm represented in performance specifications.

^ Diagnosis classification representing MCI vs. HC classification.

‡ Diagnosis classification representing AD vs. HC classification.

+ Diagnosis classification representing frontotemporal dementia subtypes (non-fluent/agrammatic variant vs. semantic variant primary progressive aphasia)

Table 3
Classification accuracy as reported by the studies using different techniques.

Algorithm (total study 97)		
Support Vector Machine (47)	Range:	65% - 100%
	Weighted Average:	86%, 95% CI [83.0, 89.6]
	Median:	89%
Non-Convolutional Artificial Neural Network (14)	Range:	71–99%
	Weighted Average:	84%, 95% CI [83.1, 84.9]
	Median:	88%
Convolutional Neural Network (22)	Range:	72–99%
	Weighted Average:	89%, 95% CI [88.5, 89.5]
	Median:	90%
Random Forest (5)	Range:	73–90%
	Weighted Average:	84%, 95% CI [81.7, 87.1]
	Median:	86%
Logistic Regression (5)	Range:	69–87%
	Weighted Average:	76%, 95% CI [72.9, 79.5]
	Median:	86%
Additional Methods (15)	Range:	65–92%
	Weighted Average:	81%, 95% CI [78.0, 84.3]
	Median:	84%

experiments attempted to improve classification by using group-wise feature weights (Linn et al., 2016), gray matter density maps (for differentiating AD and other dementias) (Möller et al., 2015), and multi-level feature characteristics for stable MCI (Long et al., 2017). Yang et al. (2017) reported an SVM with an independent component analysis (ICA) based preprocessing method on gray matter volumetric data to classify AD from HC and MCI from HC and achieved 98% and 88% accuracy, respectively.

With the establishment of biomarker-based diagnosis (Moradi et al., 2015; Frisoni et al., 2017), more recent SVM investigations also discriminated pre-clinical MCI and at-risk normal individuals using MRI combined with other predictors (Grassi, Loewenstein et al., 2018; Grassi, Perna et al., 2018; Seo et al., 2019; Sun et al., 2018; Tam et al., 2019; Zhao et al., 2019). Grassi et al. (2018) predicted 3-year AD conversion of early MCI using risk factors including MRI brain atrophy and proposed a high-performance SVM with a radial-basis kernel with ~78% accuracy. Seo et al. (2019) classified AD from HC and MCI based on the MRI volume and thickness metrics for cortical and white matter features, through non-linearly reducing MRI feature dimensions. Zheng et al. (2019) reported differentiation between clinically diagnosed vascular dementia and AD using multiple MRI features with 86% AUC.

The work by Dukart et al. (2013) used T1WI and FDG-PET to model the progression of AD and aging. Apostolova et al. (2014) stratified ApoE4 genotypes and investigated a multimodal biomarker classifier for differentiating MCI and AD with and without using hippocampal volume or CSF Aβ, t-tau, and p-tau data. Ferreira et al. (2018) studied the effect of small vessel disease on AD subtypes using MRI and CSF biomarkers and obtained an AUC of 82%. Yamashita et al. (2019) classified AD using so-called Residual Center of Mass with SVM and reported 90% and 95% accuracy with T1WI and FDG-PET, respectively.

Several studies tackled MCI using other MRI sequences besides T1WI (e.g., Dyrba et al., 2013; Haller et al., 2012. Haller et al. (2010) reported 98% sensitivity and 100% specificity using 3.0 T DTI in classifying progressive and stable MCI. Dwyer et al. (2012) investigated sex

differences in DTI based WM tracts in MCI classification. Cui et al. (2012) combined T1WI and DTI to differentiate HC and MCI and obtained a 70% performance. Dyrba et al. (2015) showed a performance of 77% classifying DTI using fractional anisotropy maps, mean diffusivity, and mode anisotropy. Dyrba et al. (2015) reported 85% accuracy, 80% AUC, 84% specificity, and 86% sensitivity, which did not increase from using only T1WI and DTI. Each of these performance measurements was exceeded by Chen et al. (2017), which combined T1WI, DTI, and DKI (e.g., 100% sensitivity). Peraza et al. (2019) used DTI to achieve an AD/HC classification accuracy of 89% and attempted to estimate the earliest detectable stages of dementia.

A few studies employed T2WI or FLAIR data. Sheng et al. (2019) identified early MCI, late MCI, and AD vs HC through a functional connectivity network. Wang et al. (2019) employed WM connectomic features with morphometric measures from T1WI, T2-FLAIR, and diffusion MRI, to compare ML models in MCI and AD classifications and predictions. Other studies employed functional MRI (e.g., rs-fMRI) to classify the diagnoses including MCI subtypes, and predicted AD conversion, and reported 91–98% accuracy (Gao Id et al., 2020; Hojjati et al., 2017; Khazaee et al., 2016; Long et al., 2016). These efforts typically involved SVM algorithm enhancements such as graph theory measures to characterize fMRI features. A study using ASL-PWI for HC and MCI classification showed 89% accuracy and 93% AUC (Collij et al., 2016).

Oliveira et al. (2015) built an MRI abnormality index using 1.5 T T1WI and a large ADNI sample. The study identified AD pathologies as deviant from the norm (i.e., healthy controls) rather than using a priori selection or feature weighting to indicate cognitive impairment. The unsupervised single-class model achieved 84% accuracy while demonstrating the potential to generalize to brain aging. Meier et al. (2012) attempted to estimate brain age based on rs-fMRI data and classified individuals as “younger adult” (18 – 35 years) or “older adult” (55 – 85 years) with 84% classification accuracy.

3.2. Non-convolutional artificial neural networks

NC-ANNs are deep learning algorithms comprised of interconnected layers of nodes, i.e., artificial neurons, capable of training and self-tuning to improve performance (Appendix A.5).

NC-ANN methods were typically used with large-scale data (e.g., ADNI) and compared with other algorithms in targeting early AD classification (Table 2B). Recent studies often report relatively high performance, likely related to novel technical developments (Cárdenas-Peña et al., 2016; Duraisamy et al., 2019; Lama et al., 2017; Mehdipour Ghazi et al., 2019; Mendoza-Léon et al., 2020; Ning et al., 2018; Zhang et al., 2019).

Yang et al. (2013) reported on a stacked network employing particle swarm optimization (an NC-ANN) with an SVM. Despite using a surprisingly small data sample, their model’s performance was comparable to a Radial basis function network involving a large 1.5 T T1WI dataset (Escudero et al., 2011), evidencing the impact of improved signal-to-noise ratio of the MRI data on increased efficiency of information retrieval. Suk et al. (2014) used a Deep Boltzmann machine to investigate both AD vs. HC diagnosis classification and MCI to AD conversion, with accuracies of 95% and 76% respectively.

Two recent studies used rs-fMRI connectivity and clinical data with NC-ANN deep learning to discriminate MCI and reported improved performance (Ju et al., 2019; Nguyen et al., 2019). Nguyen et al. (2019) employed an extreme learning machine; a multivariate pattern analysis feature selection algorithm that revealed nearly perfect model performance. Amoroso et al. (2018) used a salient skeleton network model to extract MRI features and known brain networks for AD classification and showed high performance.

As the only NC-ANN study analyzing T2WI data, Aguilar et al. (2013) compared classification performance across several machine learning algorithms and showed higher classification accuracy with a multilayer perceptron NC-ANN.

3.3. Convolution neural networks

CNNs are a type of artificial neural network that can filter data or identify the critical input features to maximize performance (Appendix A.6).

First appearing in the reviewed articles in 2017, CNNs have gained popularity quickly in MRI analysis for AD with promising performance (Table 2C) (Ghafoorian et al., 2017). The studies typically used stacked networks with CNN. The term stacked networks is used for models that employed two or more machine learning algorithms. As expected with deep learning, they involved up to tens of thousands of cases. To date, such studies use T1WI almost exclusively.

The CNN studies that combined T1WI with FDG-PET data yielded considerably different results (98% vs. 90% accuracy; Huang et al., 2019; Kruthika et al., 2019). Kruthika et al. (2019) proposed a method using a 3D capsule network and a pre-trained 3D-autoencoder and reported an accuracy up to 97.6% and 94.6% for AD/HC and MCI/AD classifications, which outperformed using only MRI or PET. Huang et al. (2019) proposed a CNN classifier to integrate multi-modality information of the hippocampal area from T1WI and FDG-PET and reported 90% accuracy for AD vs. HC and 77% for stable vs. progressive MCI.

Wang et al. (2018) tested common activation and pooling functions with an eight-layer CNN and validated with over 97% sensitivity/specificity/accuracy for AD/HC classification. Pan et al. (2020) replicated this performance by combining a CNN with ensemble learning, and brain regions most discriminable via classification of MCI to AD conversion were identified.

Basaia et al. (2019) used a more complex twelve-layer CNN and reported high sensitivity, specificity, and accuracy, each reaching 99% for AD/HC classification using separate training and validation datasets. Basheera and Sai Ram (2019) was the only CNN to employ T2WI and reached a classification accuracy of 99.2% for AD/HC.

Cohen et al. (2019) compared a 1D CNN with a feed-forward NC-ANN for multiclass classification. In addition to MRI features, CSF, ApoE4, and cognitive testing data were included in the model. The results from the two different algorithms were similar, and both performed well for discriminating clinical diagnoses (88% and 87% accuracy for the CNN and NC-ANN respectively).

Several recent studies have explored brain age by automatically extracting T1WI features using CNN and examining the relationships between brain age and chronological age in association with cognitive performance, lifestyle and genetic data, and health outcomes (Bashyam et al., 2020; Cole et al., 2017; Cole and Franke, 2017; Jonsson et al., 2019). Jonsson et al. (2019) employed 3D ResNets CNN modeling to determine brain age using healthy adults and evaluated discrepancies between predicted age and chronological age for associations with neuropsychological test performance. Bashyam et al. (2020) used DeepBrainNet, a recent CNN architecture with transfer learning, to classify neurologic diseases based on MRI brain age, and showed 86% accuracy for differentiating AD vs. HC and 70% for MCI vs. HC, comparable to models trained on patient and control datasets.

3.4. Logistic regression

LR is a type of regression model (Appendix A.9) that generates probabilities for a binary dependent variable. The method was mostly used to compare with other machine learning techniques in examining the classification performance (Table 2E). For example, Escudero et al. (2011) classified HC, MCI, and AD and suggested small differences between LR compared to SVMs, a learning tree algorithm, and an NC-ANN. Each algorithm had a performance accuracy of 84–89%.

Studies also addressed AD risk prediction. The group by Casanova et al. studied the same type of data (T1WI) for MCI to AD conversion. Their 2018 study was an extension of the 2013 work: using the same

approach, they tested the generalizability on a larger dataset to provide a score quantifying the risk of developing AD (Casanova et al., 2013, 2018). Moscoso et al. (2019) tested MCI to AD conversion based on cortex volumes and reported an accuracy rate of 86%, comparable with more complex algorithms. Wang et al. (2019) compared multiple algorithms using the same data and obtained 99% AUC, demonstrating marginal differences in algorithms.

3.5. Random forests

Random forest (RF) algorithms are based on the collective output of multiple intramural decision trees, each of which generates a prediction based on several input features (Appendix A.7).

A limited number of studies employed RF, each of which used ADNI 1.5 T T1WI (Table 2D). Gray et al. (2013) also applied FDG-PET and reported a relatively low performance compared to others with using MRI. Moore et al., (2019) used an RF algorithm with a suggested low performance of 73% accuracy. The work of Wang et al. (2019) showed a comparable performance using the same data regardless of techniques (Table 2D).

In a highly cited study, Lebedev et al. (2014) described a robust RF system for classifying AD from HC with 90% accuracy. The model was trained using the ADNI database and subsequently tested using the AddNeuroMed database. The separate testing set lends strong credibility to their reported performance metrics.

Huang et al. (2016) predicted MMSE, CDR-SOB, CDR-GLOB, and ADAS-cog scores of participants at 12-month intervals for 48 months and reported that 12-month CDR-SOB scores prediction as the most accurate ($R = 0.86$).

3.6. Additional methods used

Multiple other types of AI algorithms were applied (Table 2F), such as the regression based Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analyses (QDA) and the stacked methods which were introduced in the Technical Remarks section (Appendix A) (Franzmeier et al., 2020; Kim et al., 2019; Lee et al., 2018; Robinson et al., 2010; De Marco et al., 2017).

The study by Lee et al. (2018) used supervised learning to identify AD and MCI from HC based on cortical atrophy patterns and showed consistency of the diagnostic-specific patterns with one-year validation. Kim et al. (2019) applied T1WI based cortical thickness data to the hierarchical classification of frontotemporal dementia, AD, and subtypes using QDA, and achieved 76% overall accuracy.

Lisowska and Rekik (2019) proposed a graph theory oriented algorithm named *joint brain multiplexes pairing and mapping framework* to detect brain connectivity changes in early MCI vs. HC based on anatomical MRI and reported moderate performance. Jiang et al. (2019) employed a multi-task learning algorithm to account for correlation-aware sparse and low-rank constraints in predicting cognitive scores at various time points and confirmed featured MRI changes suggestive for AD progression.

Moradi et al. (2015) classified stable vs. progressive MCI for AD conversion by applying feature selection together with age and baseline cognitive using low-density separation and RF based supervised learning. De Marco et al. (2017) tested rs-fMRI, anatomical MRI sequences, and cognitive scores to classify MCI using LDA and QDA and showed higher performance using functional measures. Applying quantitative T1WI and DTI data of the Rotterdam study, Bouts et al. (2019) examined the generalizability of the elastic net regression model for AD and MCI classifications and reported AUCs of ~0.6.

Escudero et al. (2011) pioneered the use of ML in differentiating AD from high-risk MCI using ADNI 1.5 T T1WI for a reduction of clinical trial sample size, based on the hippocampal volume. Diciotti et al. (2012) tested the SVM and Naïve Bayesian based models with the combination of subcortical and cortical volumes, cortical thickness, and curvature features to classify mild AD vs. HC. Aguilar et al. (2013) compared the performance of supervised learning methods on AD vs. HC classification and conversion from MCI using T1WI and T2WI. Lama et al. (2017) compared several ML methods in classifying MCI and AD with greedy score-based volumetric MRI feature selection and kernel-based discrimination.

Studies targeting brain aging have also used LDA and related approaches. For example, Robinson et al. (2010) modeled structural connectivity patterns to classify older and younger adults using 83 DTI ROIs, PCA, and LDA based learning. Jiang et al. (2018) reported a fully automated pipeline named UBO Detector for clustering WM hyperintensities vs. normal tissue ROIs on T1WI, T2WI, and FLAIR, employing k-nearest neighbor-based learning that was highly agreeable with both manual tracing and rating. Beheshti et al. (2019) proposed a patch-based grading technique to estimate cortical age in cognitively normal adults and assessed its performance applying T1WI datasets and yielded a brain age highly correlated to the chronological age.

3.7. Risk of bias

The PROBAST assessment found that 56% of the studies were at high risk of bias, in terms of the three domains assessed (Fig. 3). The high overall risk was largely attributed to the Analysis domain that had the greatest risk of bias (55% rated with high risk), primarily related to an insufficient number of participants included with the outcome. The Participant domain was rated with 10% high risks, associated with inappropriate data sources or exclusion criteria, while the Outcome domain had 0% rated with high risk and 6% with insufficient information, reflecting the reporting of related features such as how the outcome was determined, were less frequently a source of bias risk across studies (Fig. 3).

4. Discussion

In this paper, we conducted a systematic review to investigate the application of common machine learning and deep learning approaches in analyzing brain MRI data for studying brain aging and AD. The search revealed 97 relevant original research articles published since January 2009 spanning 12 years. Each article was evaluated following standard methods (Page et al., 2021; Wolff et al., 2019). The study showed the unique role of machine learning and deep learning in extracting important features, uncovering hidden information, and removing noise from complex MRI data. CNN averaged the best performance for algorithm type and have been rapidly growing in popularity in recent years for AD and aging based assessment. NC-ANN trailed only slightly behind. The study also revealed several shortcomings with current AI on

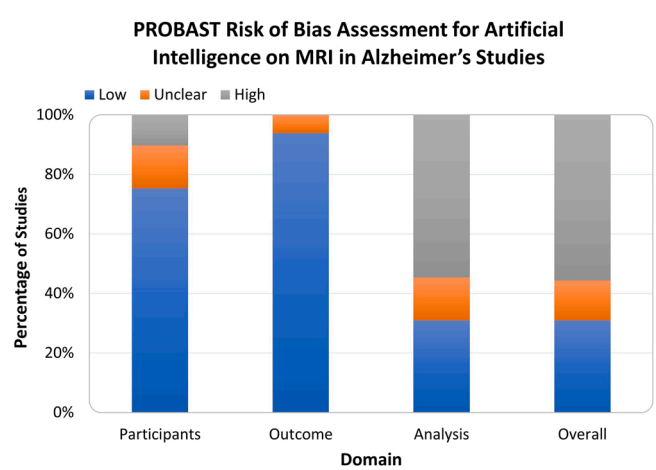

Fig. 3. PROBAST risk of bias assessment for eligible studies across Participants, Outcomes, and Analysis domains.

MRI research and suggested needed future research to advance the field.

The following caveats apply to the study. First, the search was chiefly based on PubMed indexed English publications that represent the greatest majority of the significant contributions and thus most well recognized. Although this approach has allowed the quality control of this review by focusing on peer-reviewed scientific journal contributions, less predominant publications might have been excluded, such as some conference presentations, certain e-publications, and articles of non-PubMed indexed sources. Importantly, this review excludes conference proceedings, which represent a standard communication format in computer science research targeting algorithms development. Given the multidisciplinary nature of ageing under this review, we have focused on peer-reviewed journal publications to cover AI applications on MRI in AD. Other publications focusing more on AI methods have included conference proceedings although not as updated, for supplementary information (e.g., Vieira et al., 2017). The undertaking may skew our results towards algorithms with strong performance metrics, although this is unlikely to be a major concern as the study is not meant to apprehend clinical trials or treatment outcomes. In addition, when the search terms were not shown in the title, keywords, and abstract fields that PubMed searches, some relevant articles may not have shown up through PubMed search. PubMed also employs a built-in "Automatic Term Matching" capability, which is beneficial for comprehensive searches but can limit the wildcard search. Greater coverage may be achieved by combining multiple database searches. Meanwhile, future publications are encouraged to ensure complete coverage of key components of their research in the abstract and title to facilitate wider-spread retrieval especially when reporting findings of multi-disciplinary research fields.

The search has also excluded contributions made in non-English languages. The review has focused on structural and functional MRI, and thus MR spectroscopy analyses were beyond the scope of the paper, which may deserve a separate review. No strict meta-analysis was performed due to high heterogeneity and large overlap in datasets between studies.

With respect to data accessibility, all the studies under review applied clinical diagnosis for outcome labeling, whereas a definite AD diagnosis still requires histopathologic evidence obtained from an autopsy (Alzheimer et al., 1995). Probable or possible AD diagnoses have been made applying the National Institute of Neurological and Communicative Disorders and Stroke and the Alzheimer's Disease and Related Disorders Association (NINCDS-ADRDA) criteria, based on comprehensive clinical assessments including cognitive and motor skill evaluations, medical imaging, medical history evaluation, lab testing (such as lumbar punctures), and other medical and lifestyle data. In cases where large-scale multi-center databases were used, the diagnostics have typically been verified through the effort of the databases (Jack et al., 2008; Marcus et al., 2007). Even so, the ground truth for the studies under the present review is "probable AD" or "possible AD" rather than a definitive diagnosis. Studies that did not focus on AD classification, and instead attempted to evaluate brain aging are unlikely to be biased. Expert-made lesion identification and rating may be subjective bias/error prone. Caution must be taken when they are used as ground truth for testing by the reviewed studies.

We largely evaluated the performance accuracy metric of the algorithms, which was by far the most commonly reported outcome measure used by the studies under review. Meanwhile, other metrics of algorithm performance including AUC of ROC, sensitivity, and specificity were also included whenever applicable (Table 2). Considering that AUC is a more robust evaluation metric compared to accuracy, particularly for classification tasks with unbalanced classes, we evaluated the AUC-based performance for the studies where it was reported (Figs. B2, B4, B6). It is worth noting that comparing the performance across different general types of algorithms can sometimes be nontrivial. As shown with the bias analysis, some of the studies lacked reporting the basic methodologies (e.g., regularization techniques, activation functions, inclusion/exclusion criteria) to allow for assessment. Necessary experimental design details, including the sample size and their grouping, subsets for model building and test, cross-validation methods are not always reported equally, leading to concerns of a possible bias. Regularization techniques, network pre-training, hyperparameter tuning, initialization parameters, and exact algorithm architecture (e.g., number of layers in an ANN), and datasets used can each affect performance outcomes.

This systematic review study has suggested a rapid advancement of research in this field: Of the 97 reviewed articles almost 70% were published in the last 5 years (Fig. 2, panels A, B, C), reflecting the promising future of AI on brain aging. Our work has also commented on the current status of AI on MRI analysis, to enlighten further development of the research field.

4.1. Algorithm applications

Though they vary by technical details, the studies under review have all involved the application/manipulation of computational methods for enhancing MRI feature selection and segmentation (Table 2). As shown in Fig. 2, panel A), the most common applications were classifications (n = 69, or 71%; Table 1), mostly targeting AD vs. MCI vs. HC identification. Twenty-three studies targeted prognosis for MCI to AD conversion.

A few studies targeted data quality improvement, e.g., corrections for the issues with missing and/or irregularly sampled data that are common with large AD databases (rather than a simple exclusion). Notably, an increasing number of recent studies have explored the estimation of brain age using MRI.

Appendix B provides more detailed performance comparisons across different algorithms and applications. HC vs. AD classification typically performed accurately; the weighted average accuracy was 91% across all algorithm types. However, HC vs. MCI only had a weighted average accuracy of 75%, presumably reflecting the lack of capturing early and subtle MRI differences that can be heterogeneously presented in MCI compared to that in HC. This demonstrated that the algorithms have greater difficulty picking up on small, initial brain changes, which may represent a more needed clinical application. Additionally, MCI to AD conversion had a weighted average accuracy of 79%. Identifying early warning signs of dementia or risk of disease progression for intervention and prevention will likely be a growing research focus.

Evidence from the most recent studies showed a growing trend targeting understanding and evaluating brain age gaps. Using data from 45,615 people, Kaufmann et al. (2019) compared the gaps between brain age and chronological age in several brain disorders and studied the genetic relations. Their research showed that the brains of individuals with dementia and schizophrenia were aging faster than normal. Smith et al. (2020) studied brain aging using different structural and functional changes with distinct genetic and biophysical associations, and results pointed to the heterogeneity of brain aging. Pomponio et al., (2020) reported a method to partition the brain into individual cortical and deep structures in large pooled MRI datasets for analyzing brain aging from age 3–96 years.

4.2. MRI data types

The greatest majority of the studies (n = 83, or 86%) analyzed the routinely acquired high-resolution 3D T1WI at either 1.5 T or 3.0 T (Fig. 2, panel B), while thirteen studies reported high-field (3.0 T) DTI-based analyses (14%; Table 1). Studies less frequently used other structural MRI types such as T2WI (n = 4) and FLAIR (n = 4), even though they are routinely available in clinical MRI protocols. AI studies on functional imaging were comparatively fewer, e.g., fMRI (BOLD; n = 10) and PWI using arterial spin labeling (ASL; n = 1). Several studies (n = 14) used a combination of multiple types of MRI, whereas others incorporated features from other neuroimaging (e.g., PET -

positron emission tomography), CSF (cerebrospinal fluid), genetic data (e.g., ApoE4 apolipoprotein and genome), cognitive scores, age, and sex data.

4.3. Datasets employed

The availability of open-access datasets from large-scale multi-center studies has greatly benefited the advancement of AI on AD-dementia research by lessening the time, cost, and logistics of data collection. Researchers have utilized the ADNI datasets (57% of reviewed studies) in light of its accessibility, clear information of clinical diagnosis and assessments, and longer-term follow-ups. Starting from 2018, with the development of the other large-scaled multi-center AD research initiatives (Islam and Zhang, 2018; Li et al., 2019; Mehmood et al., 2020), researchers also used OASIS (Open Access Series of Imaging studies), Sydney MAS (Memory and Ageing Study), AIBL (Australian Imaging Biomarkers and Lifestyle Study of Ageing), WHIMS-MRI (Women's Health Initiative Memory Study of Magnetic Resonance Imaging), and AddNeuroMed, while others used datasets that targeted aging and even stroke such as ICBM (International Consortium for Brain Mapping), EDSD, RUNDMC, FUTURE (Table 2). Notably, the minimum age of AD study participants was typically > 50 years to balanced diseased and control participants, whereas studies on aging often had a wider age range, which can help gauge the accumulation of the common brain changes that can begin to show on MRI as early as in the mid-20 s (Marcus et al., 2007; Petersen et al., 2010; Gu et al., 2019).

4.4. Prevailing AI algorithms

As shown in Fig. 2, panel C, most of the earliest AI-based analyses used SVM methods, which has remained most common until now, representing 48% of all the publications. More recent studies often also included SVM for comparison purposes. The initial popularity of SVM applications was consistent with the widespread SVM applications in other medical research domains and reflects the relatively early establishment of SVM algorithms (Ahmed et al., 2018; Mateos-Pérez et al., 2018). NC-ANN algorithms were also relatively common in handling MRI data (used by 14 studies, not including CNN). There were a total of 22 studies with an application of CNN or stacked networks of CNN; 13 of them were published after 2019, while the first was seen in 2017 (Table 2C; Cole et al., 2017; Li and Liu, 2018; Lin et al., 2018; Liu et al., 2018; Islam and Zhang, 2018; Wang et al., 2018; Basaia et al. 2019; Basheera and Sai Ram, 2019; Cui and Liu, 2019; Huang et al., 2019; Jain et al., 2019; Jonsson et al., 2019; Kruthika et al., 2019; Li et al., 2019; Oh et al., 2019; Spasov et al., 2019; Bashyam et al., 2020; Marzban et al., 2020; Mehmood et al., 2020; Pan et al., 2020).

The findings of this review demonstrated that CNN have exhibited more or less better performance metrics than the other major algorithm types. This observation, along with their success in other medical imaging fields, make them a promising future for clinical AD and aging applications. Meanwhile, each algorithm type has advantages and limitations. Caution must be taken when brain MRI analyses are performed using different algorithms in understanding AD and aging.

CNN and NC-ANN are powerful tools for solving regression or classification problems, however, these networks are computationally expensive to train while requiring large datasets. Their complexity makes them ideal for imaging data. The accessibility of large T1 datasets and the corresponding lack of comparably sized datasets containing other MRI scan types may be a limitation. Valuable information is contained in different scan types but may not be accessible in the quantities necessary for training deep AI networks. Further, the node weights have no easy-to-understand tangible meaning, making the final models "black boxes". Creating interpretable neural network methods is attracting increasing research attention (Mehta and Devarakonda, 2018).

SVM can also be used to categorize labeled or unlabeled data, although they typically perform less well in dealing with large datasets, especially when examples of different target classes have complex overlapping features which tend to be the case with brain imaging (Cortes and Vapnik, 1995).

RF algorithms are robust and stable, inherently more transparent than deep network algorithms, and they are relatively insensitive to the data's noise. However, they tend to perform better on tabular data than on complex imaging data. Computing the many trees and combining their outputs is costly in computational resources, making the training and decision-making processes long (Breiman, 2001).

4.5. Performances comparison

While studies so far have targeted mostly improving the classification accuracy, the reported performance for the identification of AD and MCI varied markedly from < 70% to > 99% (Table 2, Figs. B1 – B6). In other words, no one algorithm has shown a clear advantage over the rest in terms of accuracy, although on average, CNN have been associated with slightly higher mean performance (Table 3; Figs. B1 – B6), followed by SVM, NC-ANN, and RF and LR. Considerable variations exist among studies with the use of even the same general type of algorithms (Tables 2, 3). Preprocessing methods, algorithm architecture, feature identification, and other model parameters impact performance, but might not translate to improved classification accuracy. The databases used in training and testing appear to be a key factor affecting the reported performance, as different algorithms have produced comparable outcomes when applied to the same data (Wang et al., 2019). Varied data application can also be a contributor to performance, while study design should properly avoid reuse of the same IDs for model training, validation, and testing, to avoid information leaking, and potentially resulting in falsely inflated model performance. Further, studies that applied larger datasets (e.g., of 200 + participants) typically yielded comparable levels of accuracy than those with samples of just countable participants (Table 2) (e.g., Linn et al., 2016; Oliveira et al., 2010). Large sample sizes are preferable due to model optimization validity and generalizability concerns (Bryan, 2016; Mateos-Pérez et al., 2018).

The systematic review study also suggests that AI on MRI for AD research has faced several challenges currently and proposed future work to address them. First, the majority of the studies have focused on classification, although there are clearly other important focuses, on which AI can play a critical role. Instead of being limited to the identification of individuals with known clinical diagnostic labels, the predictive power of AI needs to better identify those at high risk of developing AD to provide more impactful early treatment strategies (Grassi, Perna et al., 2018). Efforts in defining the stages of AD have also been rare; a breakthrough would be key to defining an individualized care plan. Likewise, determining early risk factors is paramount. Other applications can also target improving diagnostic consistency, reducing human error, and increasing time efficiency at the point of care. AI shows promise in relating MRI-assessed brain age to AD risk (Bashyam et al., 2020), for which whole-brain outcome evaluations using calibration scores that can generate true diagnosis probabilities, or probabilities for an individual risk, e.g., using volumetric morphology measures or the atrophy and lesion index on multiple deficits (Song et al., 2013; Banerjee et al., 2018; Grajauskas et al., 2019). In doing so, regression-based techniques rather than classification-based techniques may be beneficial.

Further, despite the great endeavor towards increasing the classification performance, to date AI on MRI models that have outperformed expert evaluation as in other medical domains (Mehta and Devarakonda, 2018; Magalhaes, Mendes, and Vardasca, 2019) have yet to be realized for AD. An apparent solution lies in additional novel algorithms and innovative integrations of more algorithms optimized for targeted computations, e.g., to better extract the information. Meanwhile, it is critical to realize that the challenge with achieving consistently high performance can be deeply rooted in the complex and heterogeneous

nature of AD: as with other medical data mining problems, only when the data contain the information, can a computational method possibly extract it (Song et al., 2004). Considering that several MRI sequences are specialized in revealing different types of changes in the brain, it can be seen why models that integrated additional features from multiple types of MRI typically resulted in great performance. For instance, while reviewed publications heavily used T1WI, including T2WI or FLAIR together with T1WI better detected white matter changes.

Using information-rich data by including other MRI sequences did not only help improve feature identification with classification, but it was also critical in detecting earlier changes that may not be shown best otherwise. To this end, AI algorithms have not been sufficiently used in analyzing DTI based early white matter changes. A large-scale study demonstrated the early effects of neurovascular abnormalities in the AD brain leading to white matter hyperintensity and lacunar deficits, through capturing the trajectories of multiple factors in AD onset and progression based on the MRI and cerebral/plasma biomarkers for brain metabolism, vascular, functional, structural, and cognitive changes (Iturria-Medina et al., 2016). The constantly increasing computational power has allowed less time-consuming AI methods to tackle greater amounts of MRI data of multiple sequences.

Moreover, enhancing the transparency and interpretability of AI models can benefit the first-line care providers in understanding and trusting imaging evaluation for improved reliability of decision-making. Clinical translations may be boosted only when AI methods do not appear to be magic between images and prognostic insights. The complexities of AD and MRI necessitate more complex models, which likely is why the more straightforward random forest algorithms that are popular in several other medical fields have not been so here (Sarica et al., 2017). However, with the accuracy of AI predictions in capturing the nuances of brain aging in a black box, it is challenging, if not even impossible, to interpret the reasoning in reaching the end result. Achieving MRI-based automated decision-making will require considerable multidisciplinary collaborative research efforts involving AI, MRI data, and AD experts with shared perspectives in realizing the best of both human and machine capacities for AI solutions. For example, identifying severe AD on MRI may be easy and fast when done by a clinical expert, whereas identifying patterned subtle risks represents a challenge in which AI can help. CNN and stacked networks have started to show the greatest potential for training the machine to think like a human. Here, critical to transparency is the selection of MRI features, which demonstrates a trade-off between the complex automatic feature extraction from data and the implementation of pre-defined manual feature selection that can be expensive and time-consuming while being risky for feature loss and thus information loss (Möller et al., 2015; Long et al., 2016). Innovative methodologies that effectively combine data-driven and a-prior knowledge are most welcome: for instance, lacuna identification, brain tissue segmentation, and atrophy volumetrics and shapes can work with exploratory feature detection for transparency and meaningful interpretation.

Permitting reproducibility of research findings also presents a major challenge, posing a potential barrier for determining algorithms appropriate for clinical translation. For this reason, it is highly recommended for future publications to include technical details such as the quality, quantity, and source of information following established standards (Appendix C) (Page et al., 2021), on which performance is heavily dependent, in addition to the general algorithm. These include, but are not limited to, the independency and the size of training, validation, and testing subsets, MRI features based upon which the decision is deduced, and methods and data for cross-validation, on which risks of bias can be sufficiently understood (Appendix C) (Wolff et al., 2019). The use of broad variations can increase model generalizability (e.g., applicable to various MCI subtypes), although may lead to comparatively lower testing accuracy as specificity and sensitivity suffer in a small sample (Cui et al., 2012). Insufficient sample size can inevitably accompany deep learning, especially concerning highly complicated brain MRI, even with the continued effort of open-access data collection and sharing. Many studies have employed data augmentation to enlarge datasets and increase model exposure to variations. However, in doing so, bias can be created from testing an algorithm on the same data used to train the algorithm. Such data leakage can also be introduced when using data of individuals who have participated in multiple research projects or had MRI scans at multiple time points in a project. An algorithm performance assessment should be completed on a test dataset that is entirely independent of the training dataset, to prevent falsely higher performance and enhance generalizability (Möller et al., 2015). Most recent research has usefully attempted data simulation as reported for other research fields (Sepehri et al., 2020). Considering that a misalignment can be as high as 70–90% (Bron et al., 2015) between the postmortem neuropathological and the clinical diagnoses of AD, simulation datasets with known expert-defined brain MRI changes, e.g., hippocampal structures, may be used to help with more direct model training, when inter- and intra- rater agreement is determined/handled (Pini et al., 2016). Meanwhile, precautions must be taken to restrict data generated with augmentation or simulation to model training but not extended to model validation or testing. In this regard, public availability and sharing of datasets, including real MRI and data with augmentation and simulation, as well as algorithm codes can fundamentally help to gain reproducibility, experimental consistency, and model generalizability.

5. Conclusion

Artificial intelligence, in particular machine learning and deep learning, is continuing to evolve to increasingly play a unique role in aiding MRI analysis. However, their translation for early detection, diagnosis, and management of AD will require transparency, validity, generalizability, accuracy, and clinical salience. The increasingly larger available data and greater computing power have presented the opportunity for AI to improve, especially advanced deep learning approaches. The current AI literature has primarily focused on AD and MCI classifications, while recent algorithm development has included automating brain age assessment. Achieving identification of the nuanced MRI information of the whole brain necessary for early detection of risks for AD and brain aging warrants many additional efforts.

Author contributions

TOF conducted the literature survey, evaluation, and analysis, prepared and synthesized the results, prepared the tables and figures, and wrote the first draft of the manuscript. MG assisted with data extraction from the reviewed literature and the result summary and presentation, CCL, AZ, DP, and SGH helped with the literature search and review, bias analysis, results presentation, and manuscript drafting. RCND helped supervise the study, reviewed the results presentation, and edited the manuscript. XS conceptualized and supervised the study, obtained the research funding and trainee support, reviewed the literature survey and results evaluation and presentation, and revised the first draft of the manuscript. All authors participated in the interpretation of the results, reviewed and agreed upon submission of the final version of the paper for publication.

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Statement of Competing Interests

The authors have no competing interests concerning the preparation and publication of this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.arr.2022.101614.

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